DELIVERABLE 4

Data quality report

What the Maple Bank release really looked like, what we fixed, and what we refused to guess

Team Sharpe Shooters · CIBC Collections Hackathon · 4 October 2026

68.2M
records in 31 tables

1,052,300
people, from 3.84M source keys

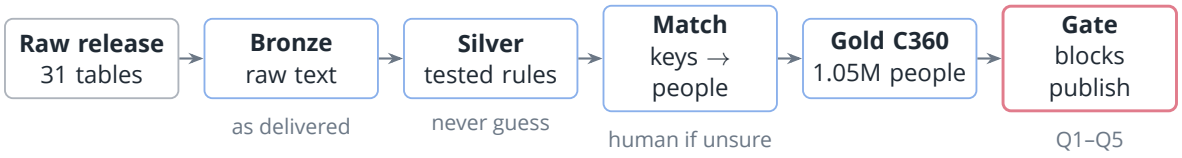
99.3%
match precision, 300 links checked by hand

PASS
contract gate, all 8 rules

The short version. The release is messy on purpose, and most of the mess is about *identity*: five systems, five customer keys, dates you can't read reliably, phones you can't use. We built the golden record by linking what we can prove first, matching the rest probabilistically, and leaving anything ambiguous empty instead of guessing. A person checked 300 random links and found 2 wrong. The contract gate passes on every rule, so the record is safe to publish downstream.

1 How we checked

Every check in this report is a query over the release or the warehouse built from it; nothing is estimated or typed in by hand. The pipeline has four stages, and each one has its own checks.



- **Land raw, clean after.** Bronze keeps every file exactly as delivered, every column as text. The release README warns that `NA` is a real outcome code and that some IDs carry leading zeros; both survive (all 228,124 `NA` contact outcomes are intact).
- **Small, tested cleaning rules** for names, dates, phones, postal codes, provinces, snapshots and the renamed column. A value a rule can't read is *never guessed*: it's left empty and written to a quarantine table with the reason.
- **Four layers of checks:** dbt tests on every model (types are enforced on gold), Soda checks generated from the data contract, the contract gate that blocks publishing, and a person for what code can't decide.

2 The release at a glance

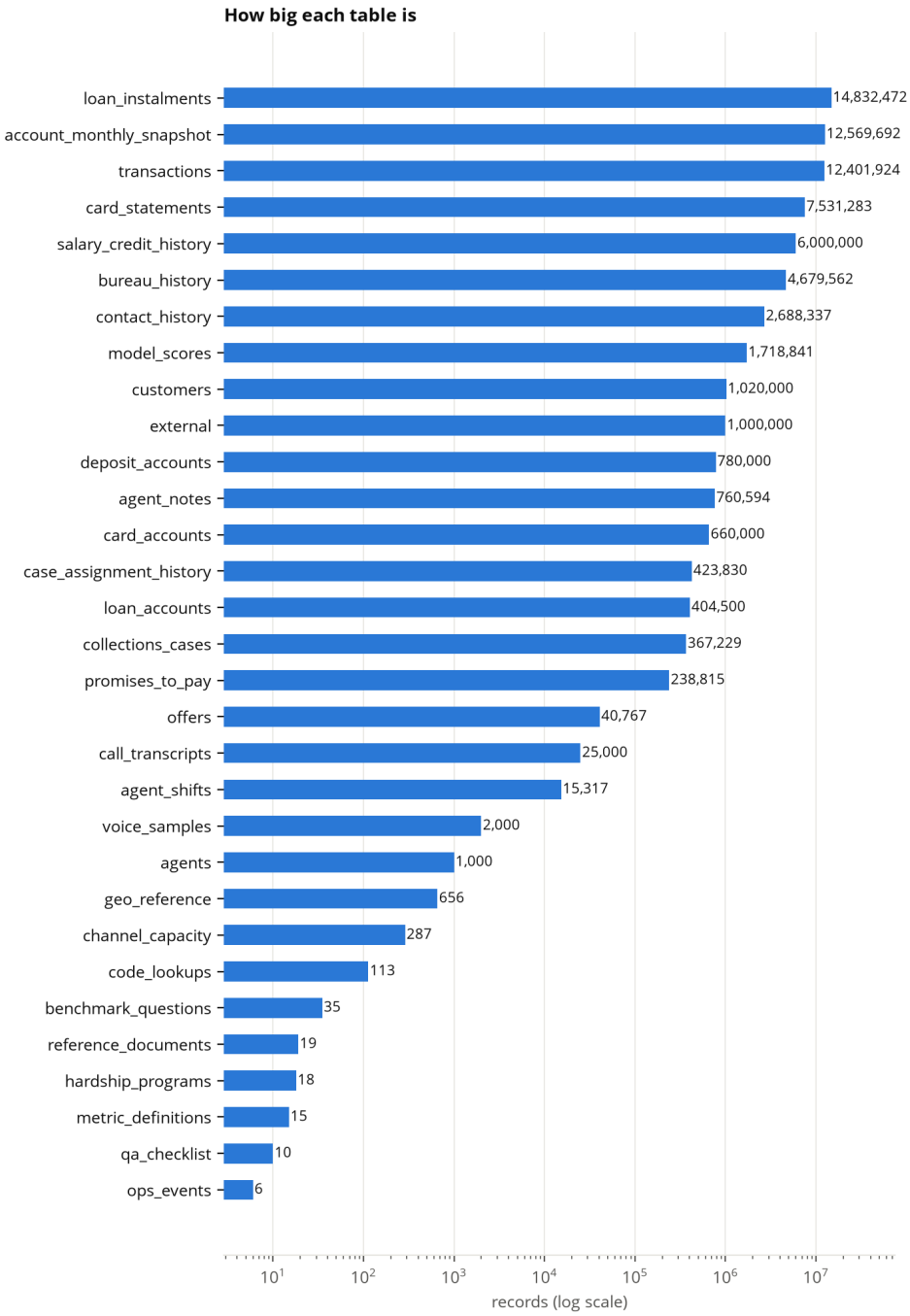


Figure 1. Records per table (log scale). Five tables hold three quarters of the rows.

What the chart says. The data is dominated by history: loan instalments, monthly account snapshots, transactions and card statements together hold 47.3M of the 68.2M records. The tables that matter most for *identity* are small by comparison (customers, cards, loans, deposits, cases: under 4M rows together), which is why matching could run on one machine. One detail already hints at the identity problem: `customers` has **1,020,000 records for 1,000,000 people**.

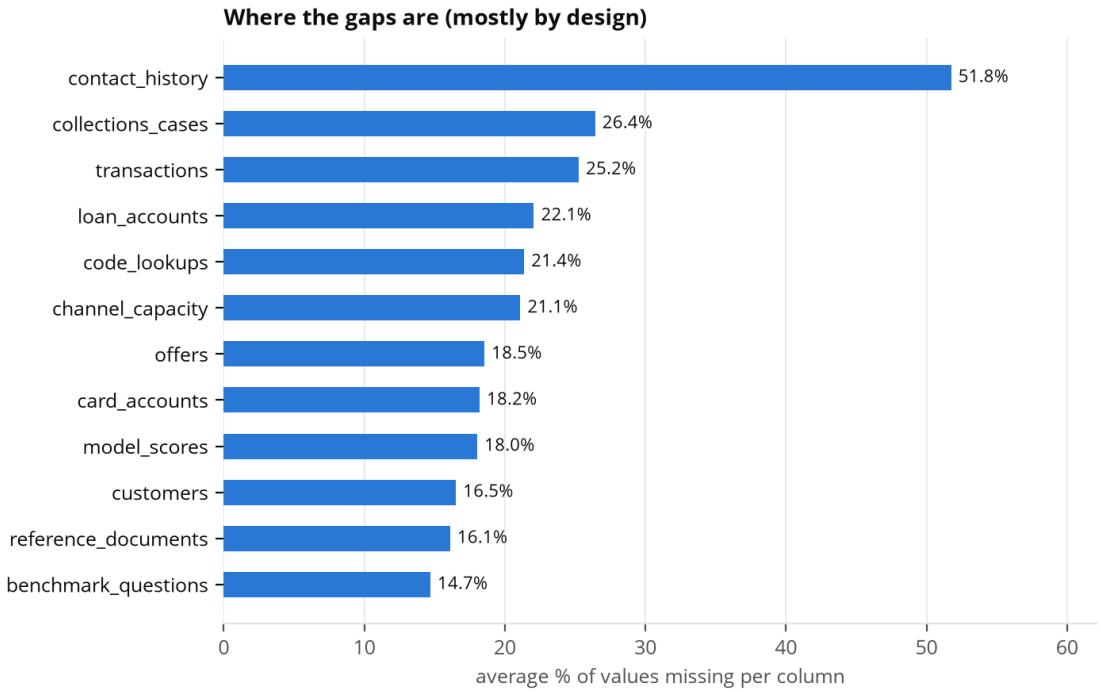


Figure 2. Average share of missing values per column, the twelve emptiest tables.

What the chart says. High numbers here are mostly *by design*, not defects. `contact_history` is 52% empty because its 128 columns describe many different event types, and each row only fills the ones for its event. A charge-off date only exists for charged-off accounts; a reversal ID only for reversed transactions. So the missing-value profile tells us the shape of the data. The actual problems are in the next section.

3 The issues, and what we did about each

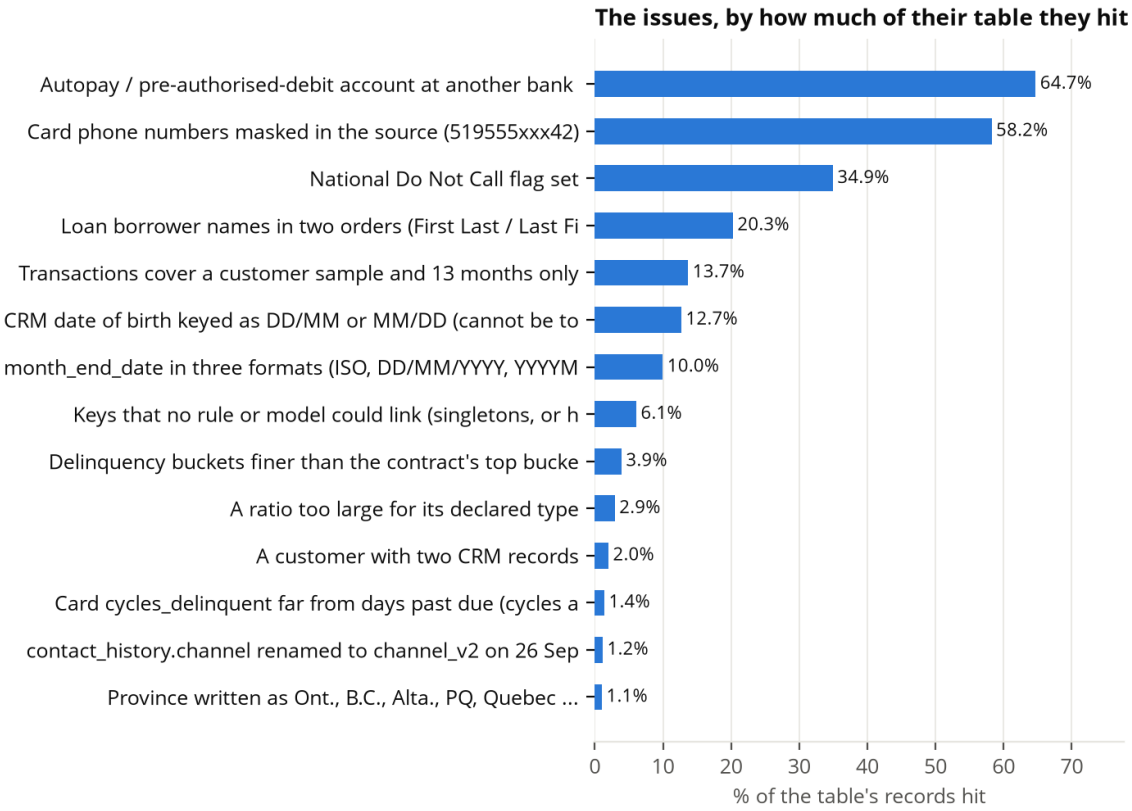


Figure 3. The issues that touch the largest share of their table.

What the chart says. The three biggest bars aren't errors in the usual sense; they're traps for a naive pipeline:

- **64.7% of autopay / PAD accounts are at other banks.** Matching on them would link strangers who happen to pay from the same external account number. We never use them to link.
- **58.3% of card phone numbers are masked** (519555xxx42). Phone is useless as a card match key, so Splink leans on name, date of birth and postal code for cards.
- **34.9% of customers carry the National Do Not Call flag.** It covers telemarketing, not collections. Treating it as a contact block would wrongly silence a third of lawful servicing contact.

The remaining issues, grouped by what they put at risk:

Schema drift. `kept map` compares the columns every raw table arrives with to the data dictionary. It flags the two `deposit_accounts` overdraft columns (described in the release README but missing from the dictionary sheet) and the transcript turn fields, which live in the JSON files rather than the CSV index.

Issue	Records	How we handle it
IDENTITY: could merge the wrong people		
DOB keyed as DD/MM or MM/DD, can't tell which	128,980 (12.7%)	Never guessed: reading all day-first misreads 16%. Left empty; other fields carry the match.
Two CRM records for one person	20,434 (2.0%)	Linked by national ID, CRM's own duplicate pointer, or Splink.
Loan names in two orders (First Last / Last First)	82,028 (20.3%)	The token CRM uses more often as a surname is taken as the surname.
CRM phones cut to 9 digits	4,936 (0.5%)	Not a valid number: left empty, quarantined.
PARSING: values that silently disappear		
`month_end_date` in three formats	1,255,152 (10.0%)	A plain date cast drops them; silver reads all three.
Province as Ont., B.C., PQ, Quebec…	10,689 (1.1%)	Mapped to the two-letter code before gold.
A ratio too large for its declared type	13,969 (2.9%)	Silver uses a wider decimal; the values look like a unit problem and aren't used.
Notes with lost characters (U+FFFD)	4,636 (0.6%)	Unrecoverable; the extractor reads the surrounding words.
STRUCTURE: schema and grain		
`channel` renamed `channel_v2` on 26 Sep	31,428 (1.2%)	Coalesced back into one column.
Deleted CRM records still in the extract	2,115 (0.2%)	Left out of silver; listed in quarantine.
Buckets finer than the contract's top bucket	14,409 (3.9%)	Silver keeps the label; gold derives the bucket from days past due.
Card cycles far from days past due	673 (1.4%)	Gold and features use days past due, never cycles.
Duplicate primary keys	0	Checked on every load; none found.
COVERAGE: data that only exists for some customers		
Transactions: a sample of customers, 13 months	106,914 (13.7%)	Features from them are empty, never 0, outside the sample.
Keys nothing could link (singletons or held apart)	64,092 (6.1%)	Each keeps its own golden ID with status `unlinked`.

Table 1. Issues by the risk they carry. Percentages are of the affected table.

4 Do the tables actually join?

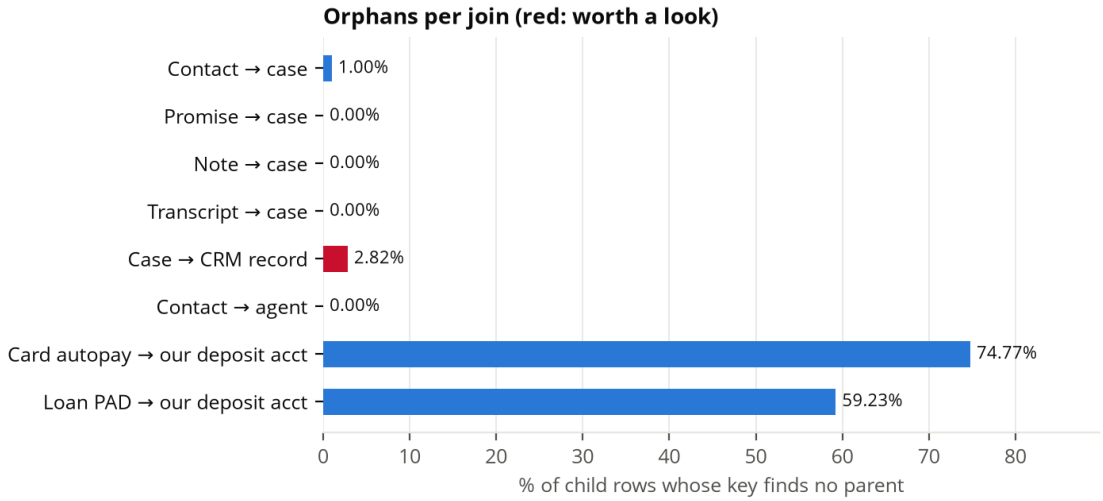


Figure 4. Share of child rows whose key finds no parent row. Red: worth a look.

What the chart says. Notes, transcripts, promises and agent references join perfectly. Two joins deserve a closer look:

- **Case → CRM record: 10,365 cases (2.82%) point at a CRM ID that isn't in `customers`.** This was the one real surprise. Those cases aren't lost: their card, loan and deposit keys, plus Splink, still put them on a person, and the gate checks that every *open* case resolves to a golden ID.
- **Contact → case: 26,864 orphans, exactly 1.00%,** right at the tolerance of the release's own example contract. They're kept and counted but join to nothing.

The autopay and PAD bars are tall on purpose: most of those accounts are at other banks (§3).

5 Identity resolution

This is the step that can actually hurt a customer: a wrong merge means calling someone about a stranger's debt. So it gets the most checks.

5.1 How keys become people

Two passes. First, **exact rules** link keys we can prove: the case's copy of the CRM ID, autopay/PAD to our own deposit account, the same national-ID hash, CRM's own duplicate pointer, and the core-banking customer number (CIF) shared by deposits and the bureau file. Then **Splink** (Fellegi–Sunter matching on name, date of birth, phone and postal code) scores what's left. A score of 0.95 or more links automatically, 0.80–0.95 goes to a data steward, and anything lower stays apart. A cluster with two national IDs or more than 12 keys is never merged automatically.

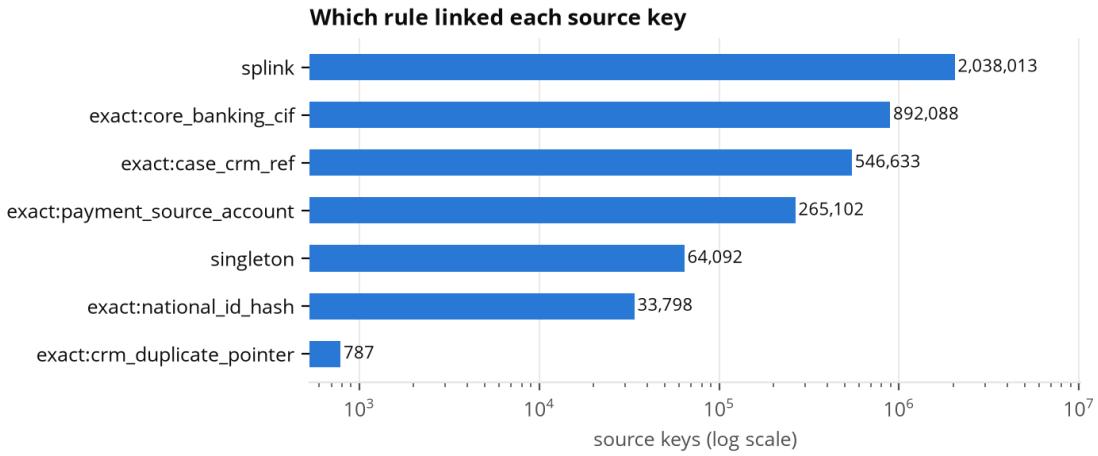


Figure 5. Source keys by the rule that linked them (log scale).

What the chart says. Splink does most of the work (2.04M keys, 53%), but the exact rules carry the backbone: the CIF rule alone links 892K keys, and it wasn't in our original design; we found it in the release README and checked that name, DOB and postal code agree on 99.7–100% of those pairs. The national ID, the field you'd expect to be the strongest, links only 33.8K keys, because most records don't carry one.

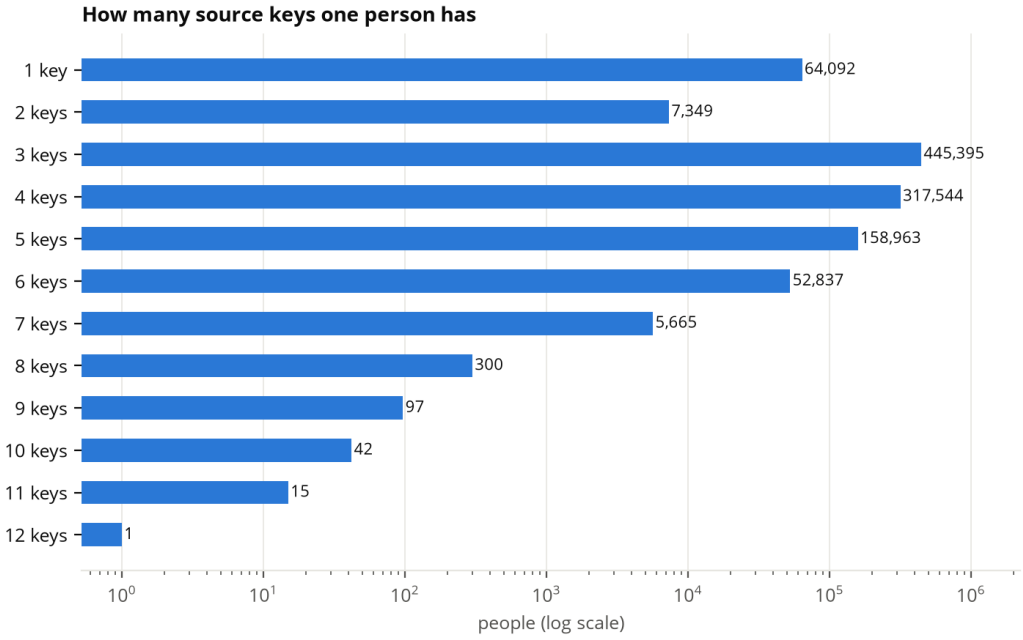


Figure 6. How many source keys one person has (log scale).

What the chart says. The typical person has **three to five keys** (921K of 1.05M people), which matches a CRM record plus one or two products plus a case. 64,092 keys stay single. The tail stops at 12 keys by design; seven larger clusters were held for review rather than merged.

Splink is decisive. Every key Splink auto-linked scored 0.98 or higher, and only 251 pairs fell into the 0.80–0.95 band that goes to a steward. The model rarely sits on the fence, which makes the human review sample (next) the main check on whether it's *right*, not just confident.

5.2 The human check

The organisers gave no ground-truth IDs, so a person reviewed 300 randomly sampled predicted links, sorted into tiers by how many identifiers agree (276 clear, 14 check, 10 look closely).

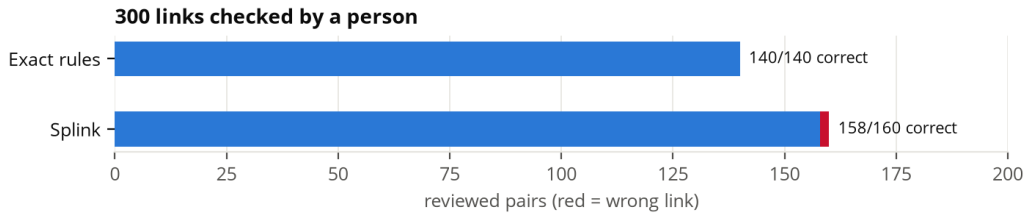


Figure 7. 300 reviewed links by the pass that made them. Red marks a wrong link.

Pass	Reviewed	Correct	Precision
Exact rules	140	140	100.0%
Splink	160	158	98.8%
All	**300**	**298**	**99.3%** (target 98%)

Table 2. Match precision by pass.

> **Why the two errors matter**
> Both wrong links (MP-238, MP-239) are Splink pairs where everything matched *except the surname*, and both scored the maximum, 1.0. No threshold would have caught them. That's the argument for keeping a reviewed sample in every release, and for the rule that a cluster holding two national IDs is never merged.

Recall, by injection. Precision says the links we make are right; recall says we make the links we should. The CRM's own duplicate pointer can't measure it (the matcher uses it as an exact rule), so we planted 200 copies of real customers with no national ID and no pointer, each damaged one way (exact copy, surname typo, no phone, new postal code, first initial only), and re-ran the matcher. **198 of 200 (99%) merged with their original**; the 2 misses stayed a person of their own (no wrong merges), and 2 of 50,036 other clusters shifted. Details: `docs/reports/identity_recall.md`.

5.3 What the golden record holds

People in `gold.c360_customer`	1,052,300
of which single keys nothing could link (`unlinked`)	64,092
`do_not_contact` set (any linked record said so)	37,042
`collections_hold` set (insolvency, bankruptcy or deceased)	23,593
Hardship confirmed / possible	32,306 / 28,061
Golden IDs merged or split since the previous run	0 / 0

Table 3. The published golden record.

Contact flags take the *most restrictive* value across every linked record: if any one of a person's records says "do not call", the person isn't called.

6 The contract and the gate

The golden record is a data product with a contract (`contracts/c360_customer.odcs.yaml`). We started from the release's example contract, DC-COLL-001, and extended it to the Open Data Contract Standard:

DC-COLL-001 asks for	Where it lives in our contract
Owner	`team`: business owner, technical owner, data steward
Refresh	`slaProperties`: daily, published by 06:00 ET; consent and holds within 15 minutes
Quality rules	`quality`: Q1–Q5 with warn and block thresholds
Allowed uses	`description.usage`: servicing, the NBA with human release, the Ask, hardship identification
Not allowed	`description.limitations`: marketing, credit or pricing, protected attributes, bulk export

Table 4. The organisers' example contract, mapped to ours.

Rule	What it checks (result)	Status
Q1 unique IDs	1,052,300 golden IDs, unique; every active source key maps to exactly one	**PASS**
Q2 linkage	98.66% of account keys linked ($\geq$ 98%); every open case resolves	**PASS**
Q3 match safety	every auto link $\geq$ 0.95; no cluster holds two national IDs	**PASS**
Q3 review precision	99.3% on 300 reviewed links ($\geq$ 98%)	**PASS**
Q4 contact safety	0 people with a source cease or hold flag missing from gold	**PASS**
Q5 freshness	published within 26 hours of the extract	**PASS**
Q5 row count	within 10% of the previous load (+0.0%)	**PASS**
Hardship from text	785,594 extractions loaded; 11,464 people rest on them	**PASS**

Table 5. The gate on the published record. Any block-level failure stops publishing.

7 Beyond the golden record

- **Text.** All 760,594 notes and 25,000 transcripts are redacted (English and French spaCy, Canadian SIN and postal codes, mangled cards and dates of birth) before any model reads them. A full scan afterwards finds **0** leftover identifiers, and the LLM gateway refuses anything that isn't redacted.
- **Labels: public only.** Text models are tuned and tested only on the release's 500 public labels per source, 30% held out. The remaining labels belong to the organisers.
- **Features.** Training and live values agree on all 56,000 compared cells. The drift report covers how tonight's features compare with training.

8 What we can't measure

- **True match accuracy.** There's no ground truth. The human review and the exact-rule cross-check (Splink re-finds 99.9% of the links the exact rules prove) are the evidence.
- **Monthly roll rates.** The account snapshot only has quarter-ends (Dec 2023 to Jun 2026), so roll rates are quarterly.
- **Generated fields.** The release README says some fields were set from generation rules; we take them as given.

9 If we owned the source systems

1. Carry one customer ID, or at least the CIF, in CRM: 892K of our links depend on a number only deposits and the bureau hold.
2. Force one date-of-birth format at entry: 12.7% of CRM dates of birth are ambiguous today.
3. Stop masking card phone numbers in the extract.
4. Publish schema changes (`channel_v2`, the overdraft columns) in the dictionary before they ship.
5. Keep a 300-link human review per release: it caught the only two bad links.

Numbers `scripts/quality_report_full.py` queries the release, the warehouse and the identity crosswalk, and caches everything in `data/exports/reports/quality_data.json`.

Rebuild `uv run python scripts/reportlib.py quality_report`